AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 19 · STUDENT LESSON

User Need and AI Contribution

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Name the need. Compare the baseline. Justify the contribution.

Design a useful bounded helper before adding AI features.

Keep one habit: Need-to-test contract → glossary case → two design flows → evidence plan → independent equipment scope.

MAIN LESSON · SECTION 1

What does the learner need help doing?

> Choose AI only for a defined contribution that can be compared with a simpler alternative.

Your learning goals

I can explain the **mechanism** and terminology of capstone **requirement**s and corpus.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Recall evidence and a baseline

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Recall Retrieval Augmented Generation: a fluent answer must still be checked against the selected **evidence**. Before designing a helper, state the user's task and the simpler way of doing it.

Your end goal

By the end, I can explain **user need** and **ai contribution** using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Retrieval Augmented Generation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Verify Numbers and Dates. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Multimodal **Evidence**. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 19: **User**, **Need**, and **AI Contribution**

AI should solve a defined **user** problem better than a **simpler alternative**.



CREATOR MISSION

Compare AI and non-AI designs for a study **need**.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Name the need before choosing the tool

> A requirements plan is a proposal, not evidence of user benefit.

Understand user need and baseline

Start with a user, a task and a difficulty, rather than a feature list. Example: a pupil needs to locate an unfamiliar word in an approved glossary and understand its meaning. A baseline is a simpler way to meet the same need, such as direct glossary search. AI might help restate an entry, but that is a proposed benefit until evaluated.

If the baseline already works well, an AI feature needs a clear additional reason.

Connect the need to a bounded capstone

A grounded-assistant capstone begins with bounded **requirement**s and a versioned permitted corpus. A baseline can be a manual lookup **process** using the same **evidence**.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use **user**, **need**, and **ai contribution** to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

Turn the need into a checkable service boundary

> Specify evidence, output, fallback and comparison before implementation.

The concrete need-to-test contract

Requirement	Concrete glossary example
User and task	Pupil locating an approved word meaning
Permitted evidence	G3-v1 only: habitat = the place an organism lives
Baseline	Find and read the glossary entry directly
Possible AI contribution	Locate and clearly restate the same supported meaning with its source
Boundary	No answer for an absent entry; report not in this glossary

Requirement	Concrete glossary example
Evaluation plan	Compare supported completion, time, errors and help used on fresh matched tasks
Decision	Keep a feature only if observed value justifies its extra errors, effort and cost

Preserve requirements and evaluation separation

Specify the **user** question, source authority, access, **output**/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation. Keep teacher material out of student retrieval.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the deciding check

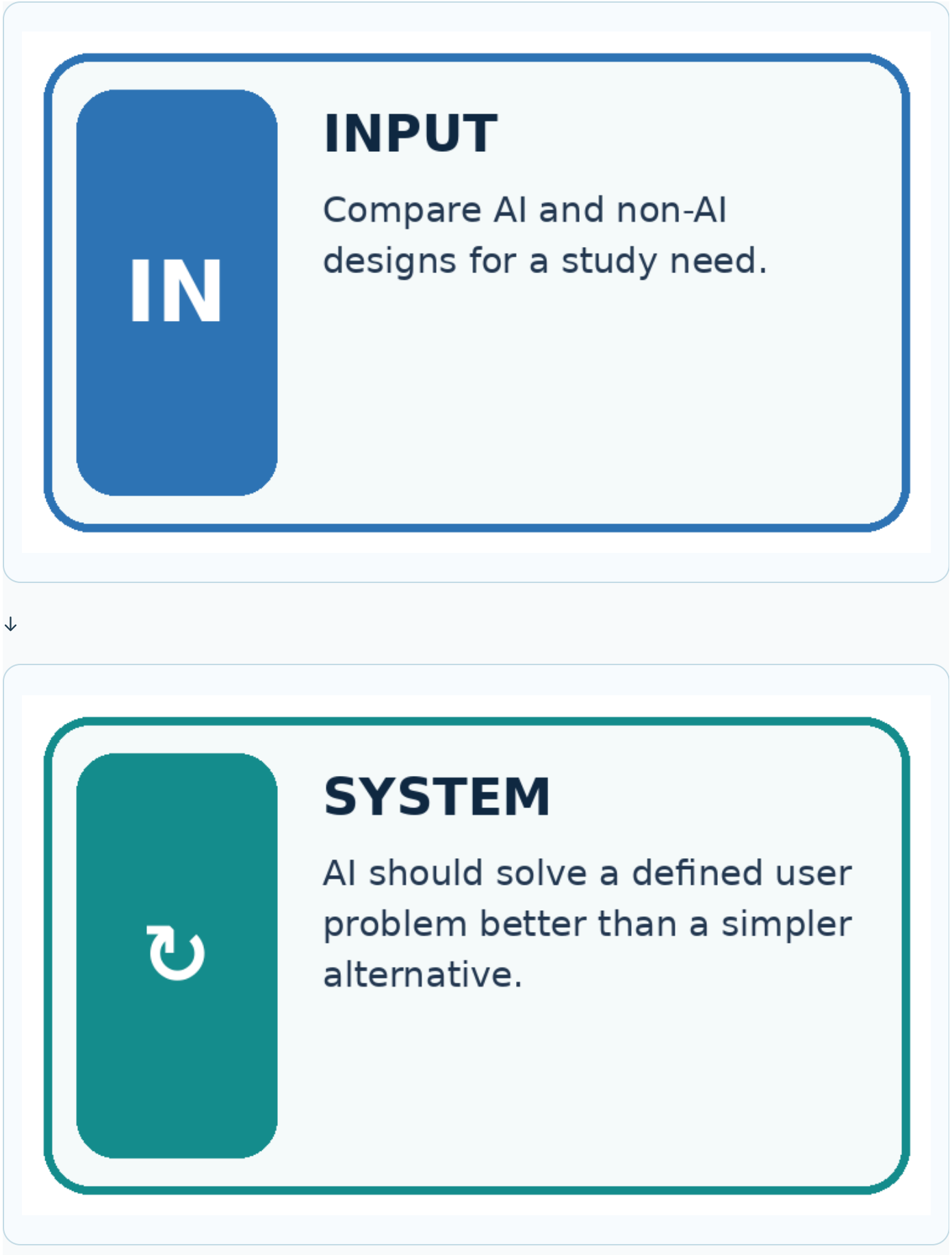
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

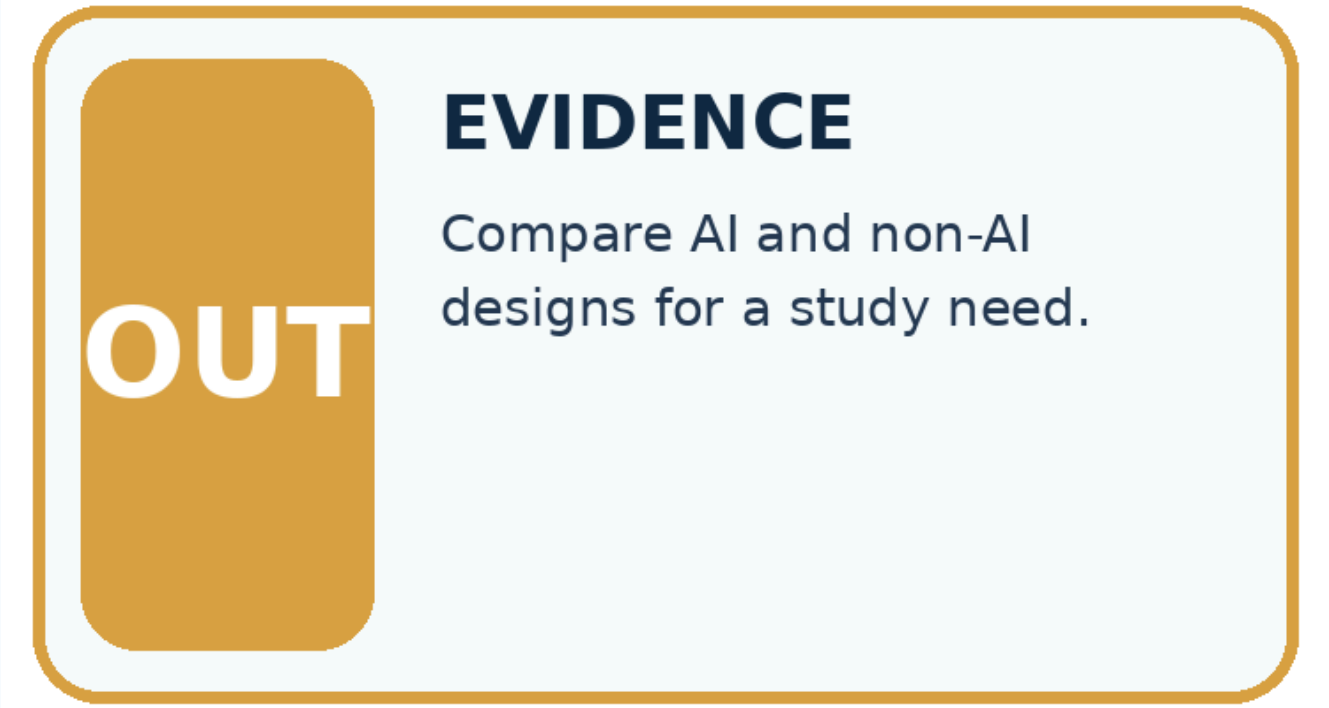
VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Compare AI and non-AI designs for a study need.
↓
SYSTEM
AI should solve a defined user problem better than a simpler alternative.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Capstone requirements and corpus — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Student room assistant corpus contains public register P, teacher-only guide T and current approval A. It must report approved room capacity or needs_review.	Specify the user question, source authority, access, output/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation. Keep teacher material out of student retrieval.	Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity. Reserve complete, missing and conflicting cases before development.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

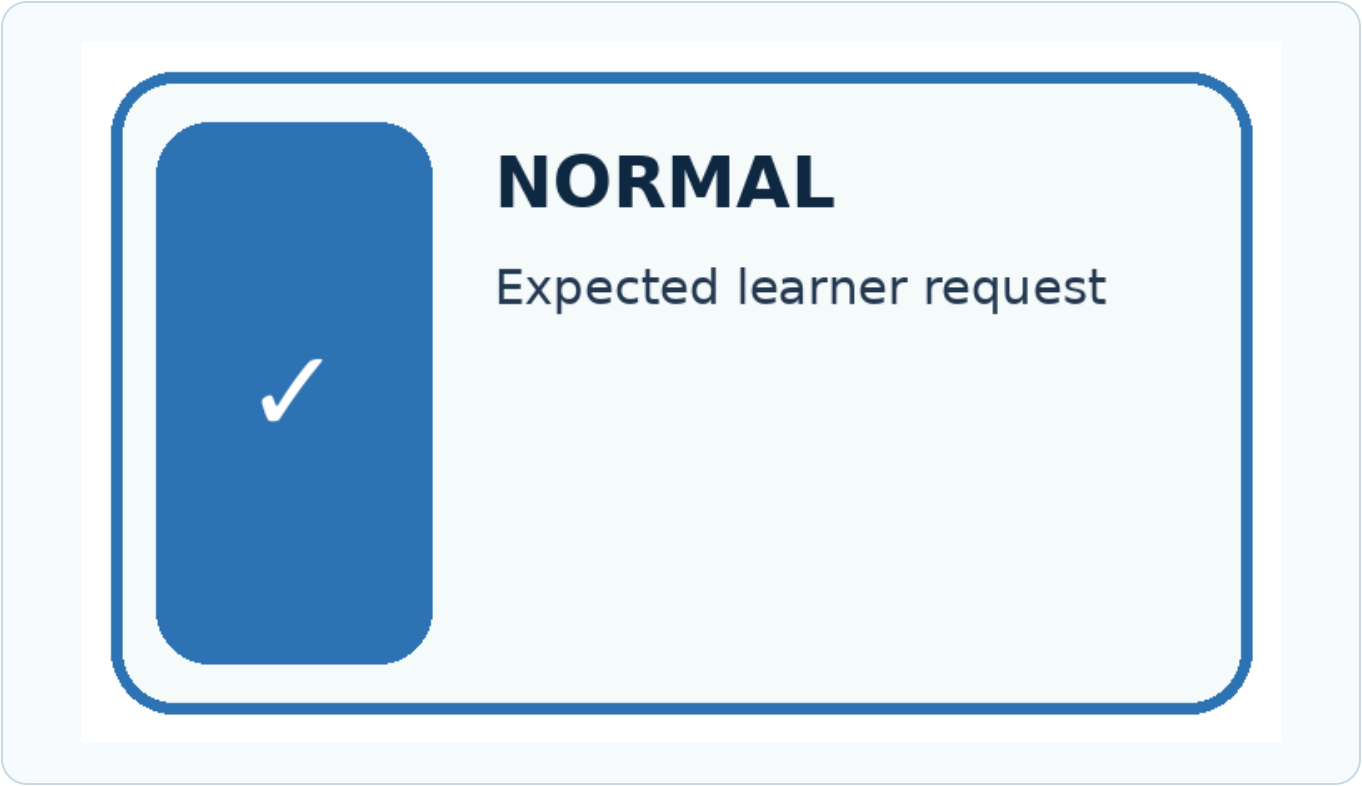
Connect the picture and the explanation

MAIN LESSON · SECTION 4

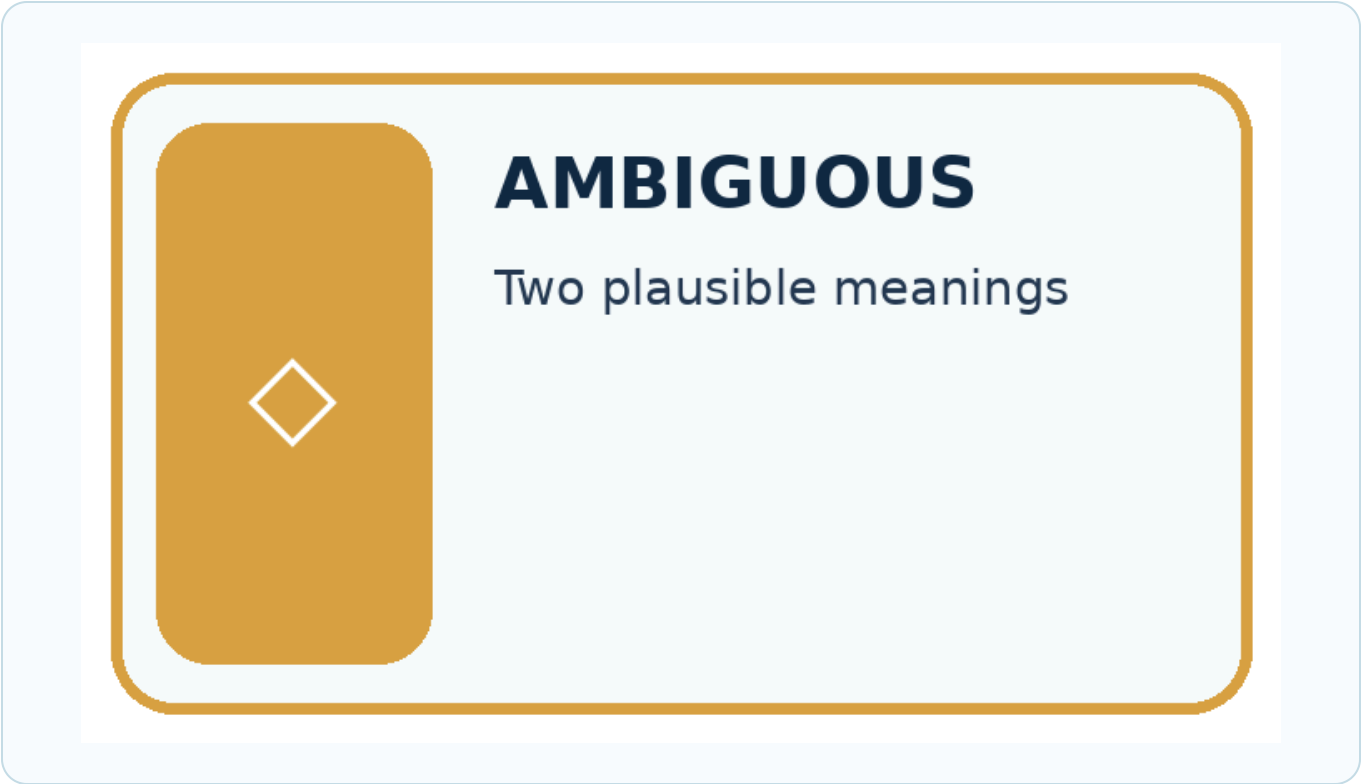
Compare useful help with an unbounded promise

> A chatbot that answers everything is not a specification for a glossary need.

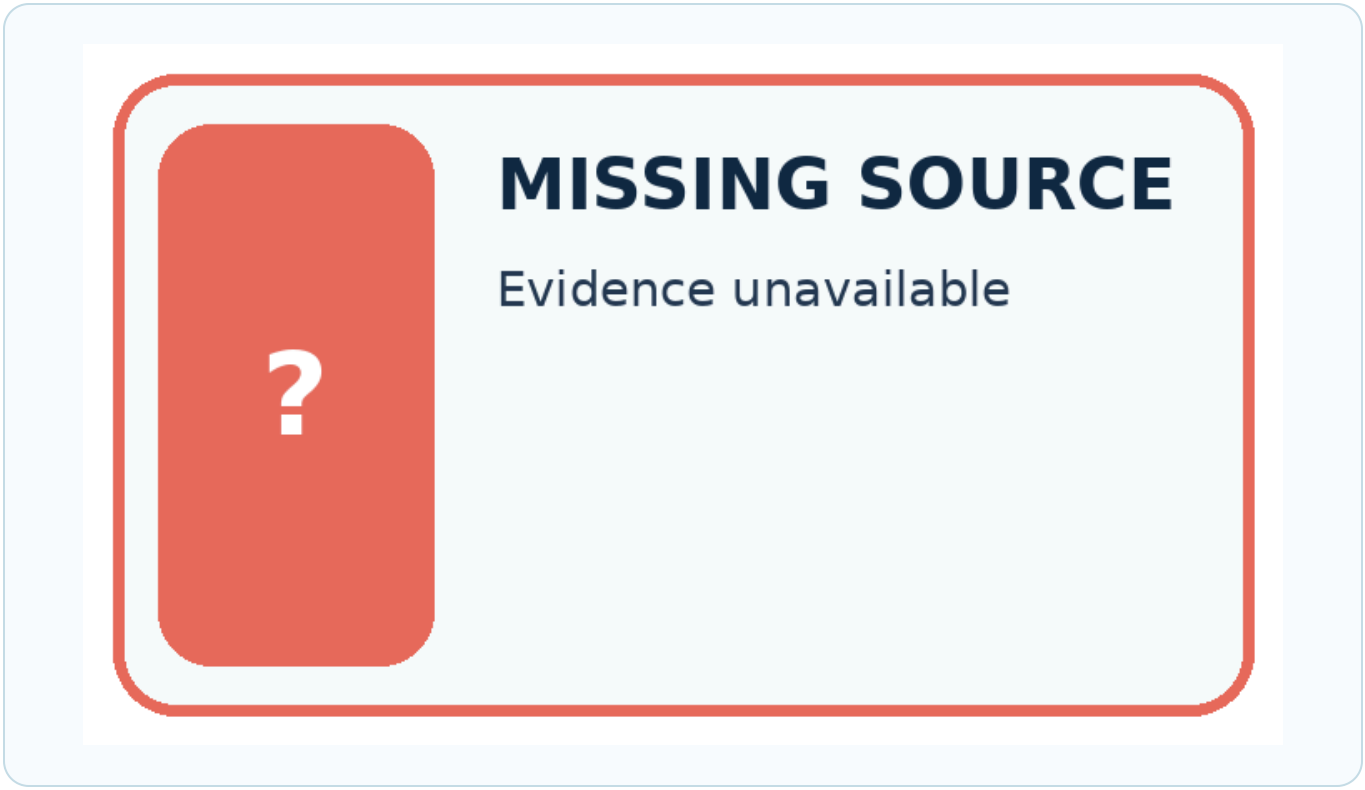
Four pictures with a purpose



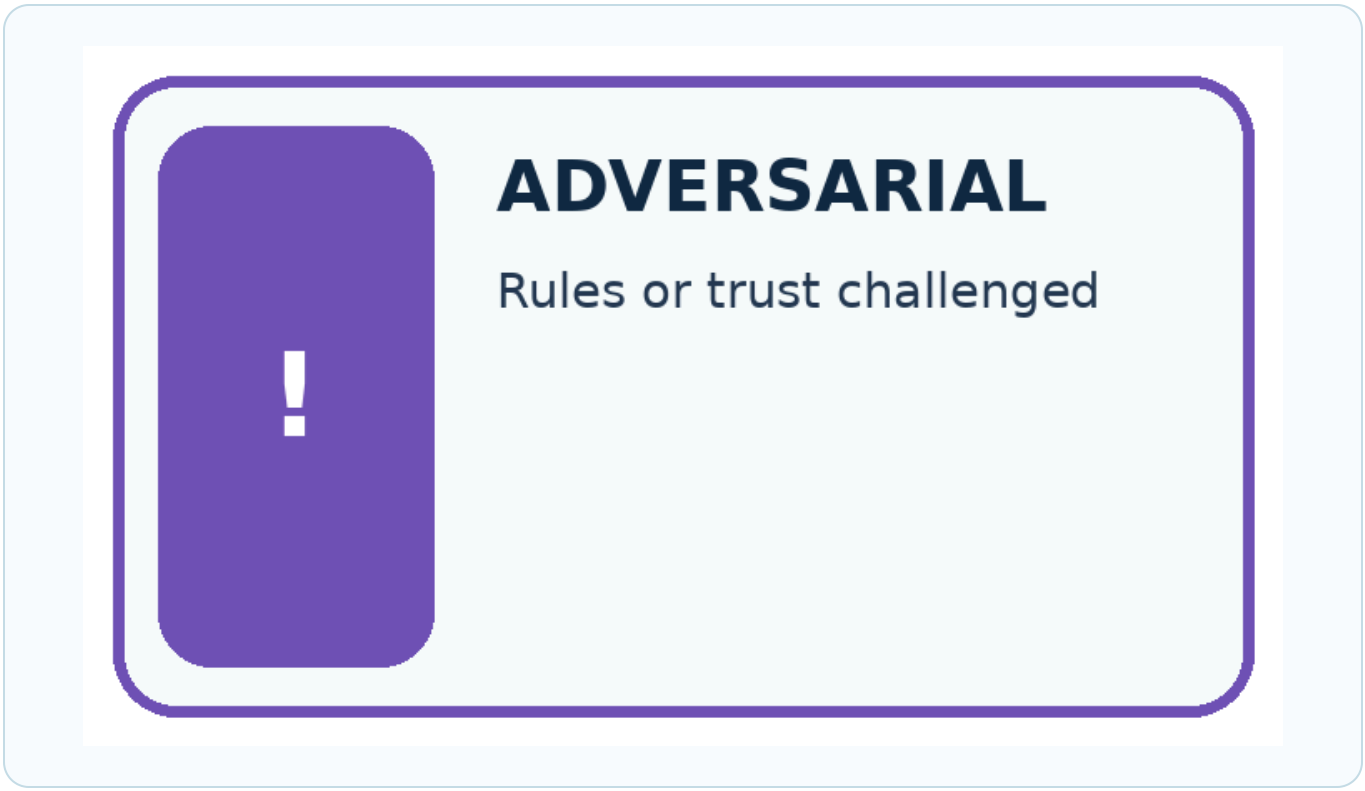
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: The pupil asks for photosynthesis, which G3 does not contain.

▼ Check after predicting

Prediction: The assistant reports that the definition is unavailable in its approved glossary.

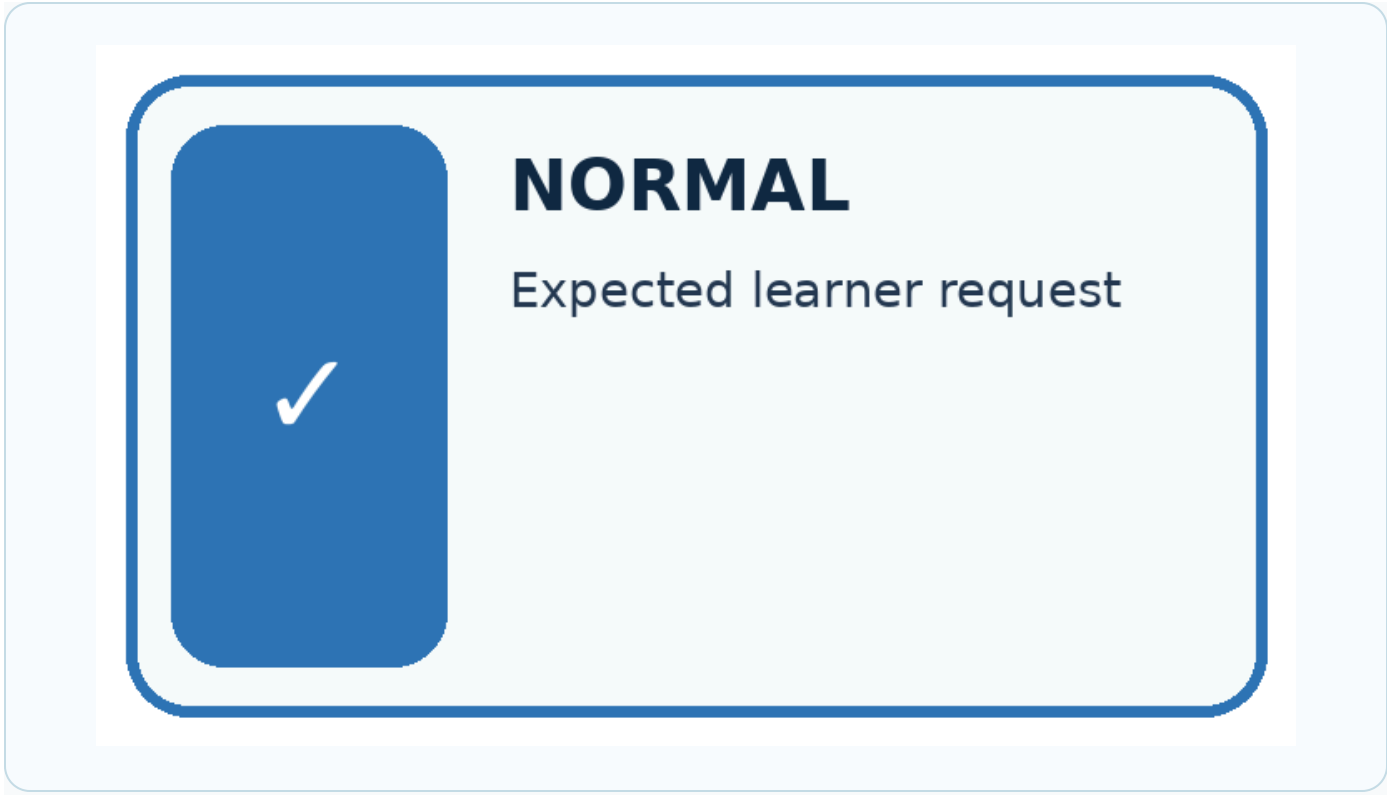
Reason: A helpful interface cannot expand its authorised **evidence** by guessing.

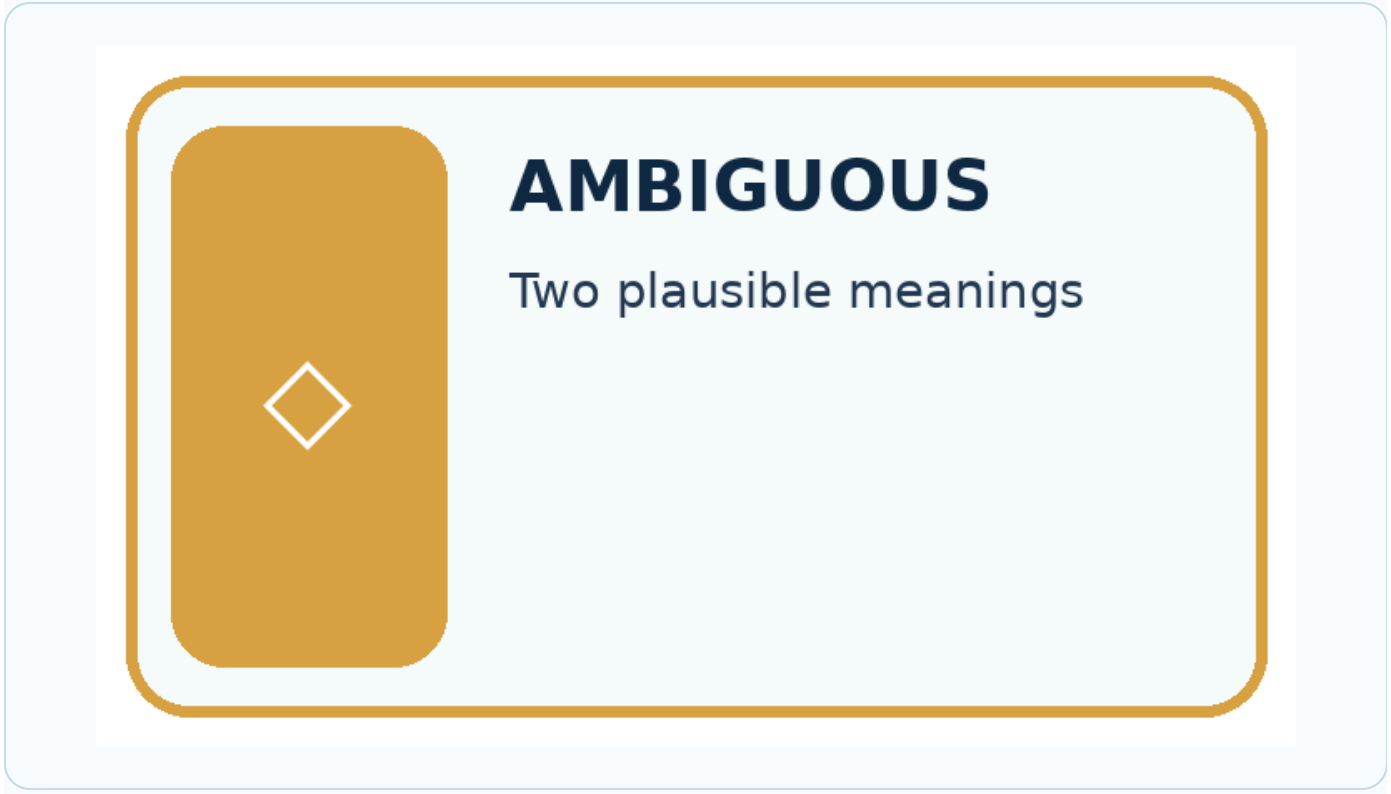
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace the glossary helper’s bounded contribution

> Separate the user-need example from room-corpus planning practice.

Read the complete glossary scope

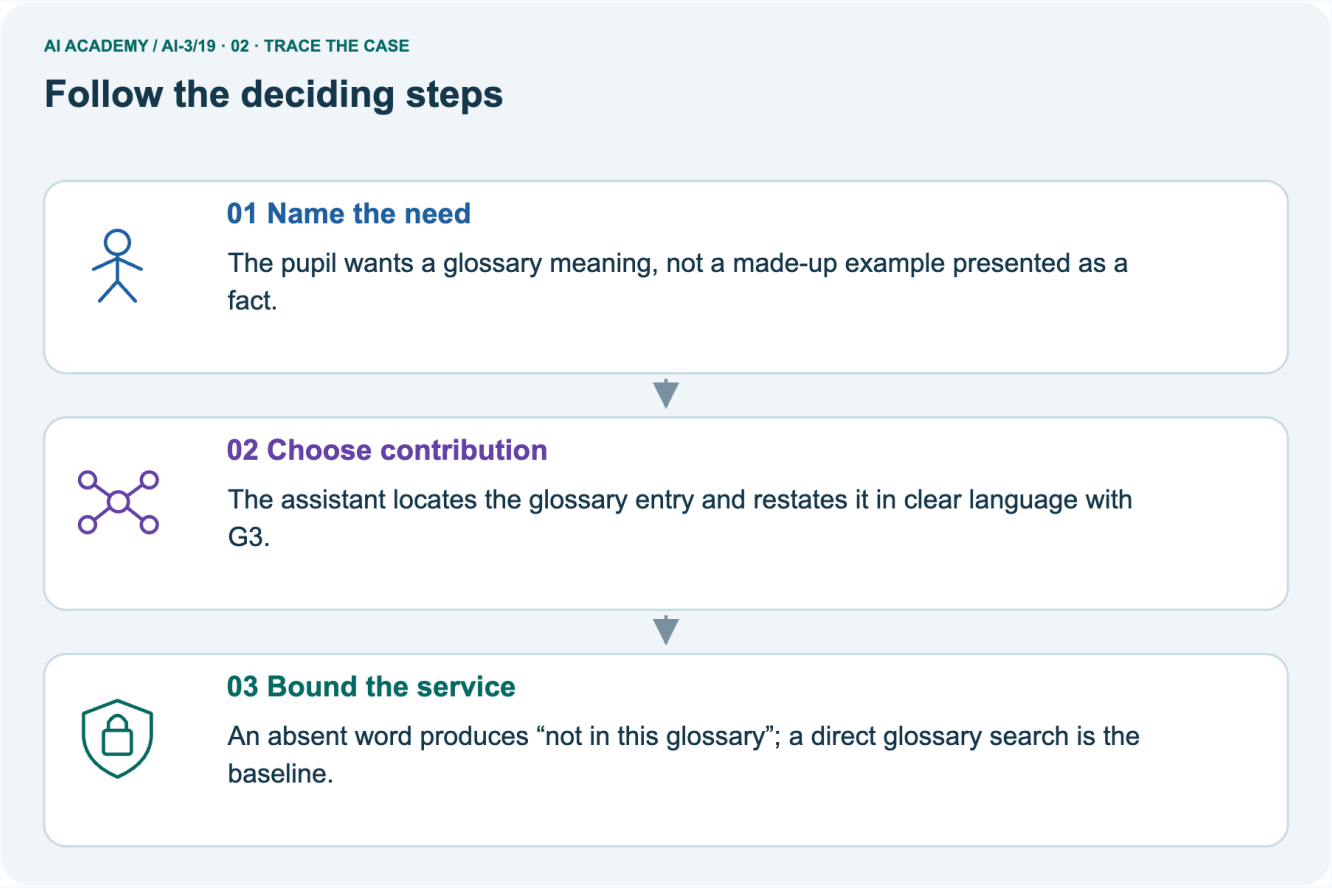
Define a useful reading helper before choosing its AI features.

A pupil **need**s the meaning of unfamiliar words in an approved glossary. G3 defines habitat as the place an organism lives; no other sources are allowed.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

For this complete toy glossary, G3-v1 contains only habitat: the place an organism lives. Photosynthesis is absent. Both direct lookup and the proposed grounded helper must report that absence. The proposed helper may restate the supported meaning; it cannot add unauthorised factual content. No usability study, model run or measured learning gain is supplied.

Trace need, contribution and fallback



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Name the need	The pupil wants a glossary meaning, not a made-up example presented as a fact.
2. Choose contribution	The assistant locates the glossary entry and restates it in clear language with G3.
3. Bound the service	An absent word produces “not in this glossary”; a direct glossary search is the baseline.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

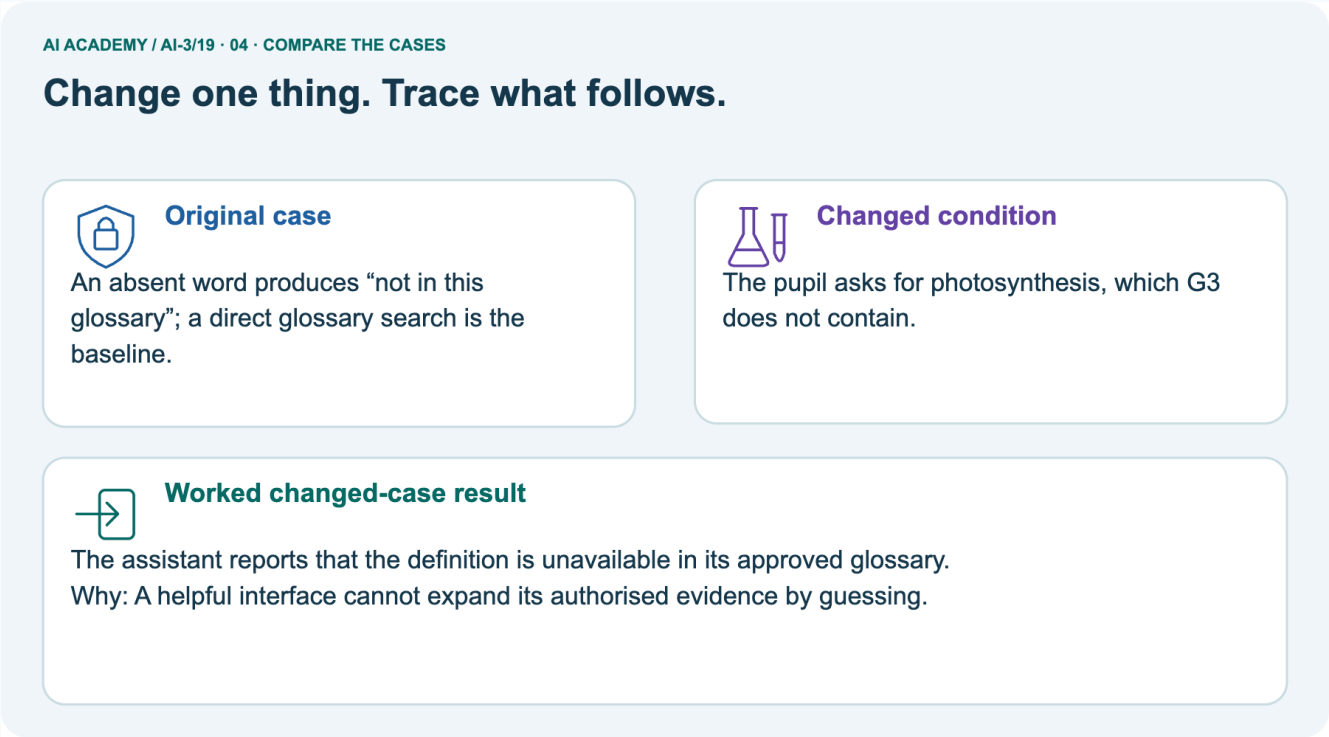
Read the picture in order. An absent word produces “not in this glossary”; a direct glossary search is the baseline. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict asking for an absent word

- Change one thing: The pupil asks for photosynthesis, which G3 does not contain. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Name the unmeasured benefit

- A **requirement**s plan identifies intended value; it does not demonstrate actual pupil benefit.

- Read the labelled picture: What non-AI baseline would reveal whether this helper adds value? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

▼ Separate practice · Room-assistant corpus plan

Student room assistant corpus contains public register P, teacher-only guide T and current approval A. It must report approved room capacity or **need**s_review.

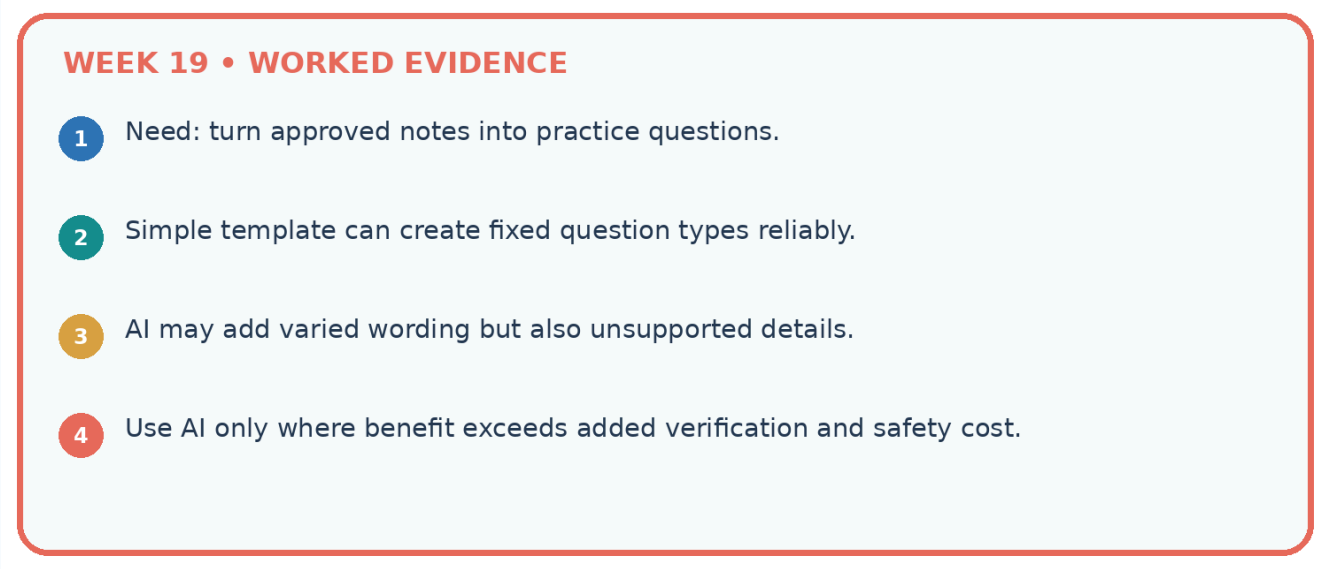
Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity. Reserve complete, missing and conflicting cases before development.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. **Need**: turn approved notes into practice questions.

2. Simple template can create fixed question types reliably.

3. AI may add varied wording but also unsupported details.

4. Use AI only where benefit exceeds added verification and safety cost.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

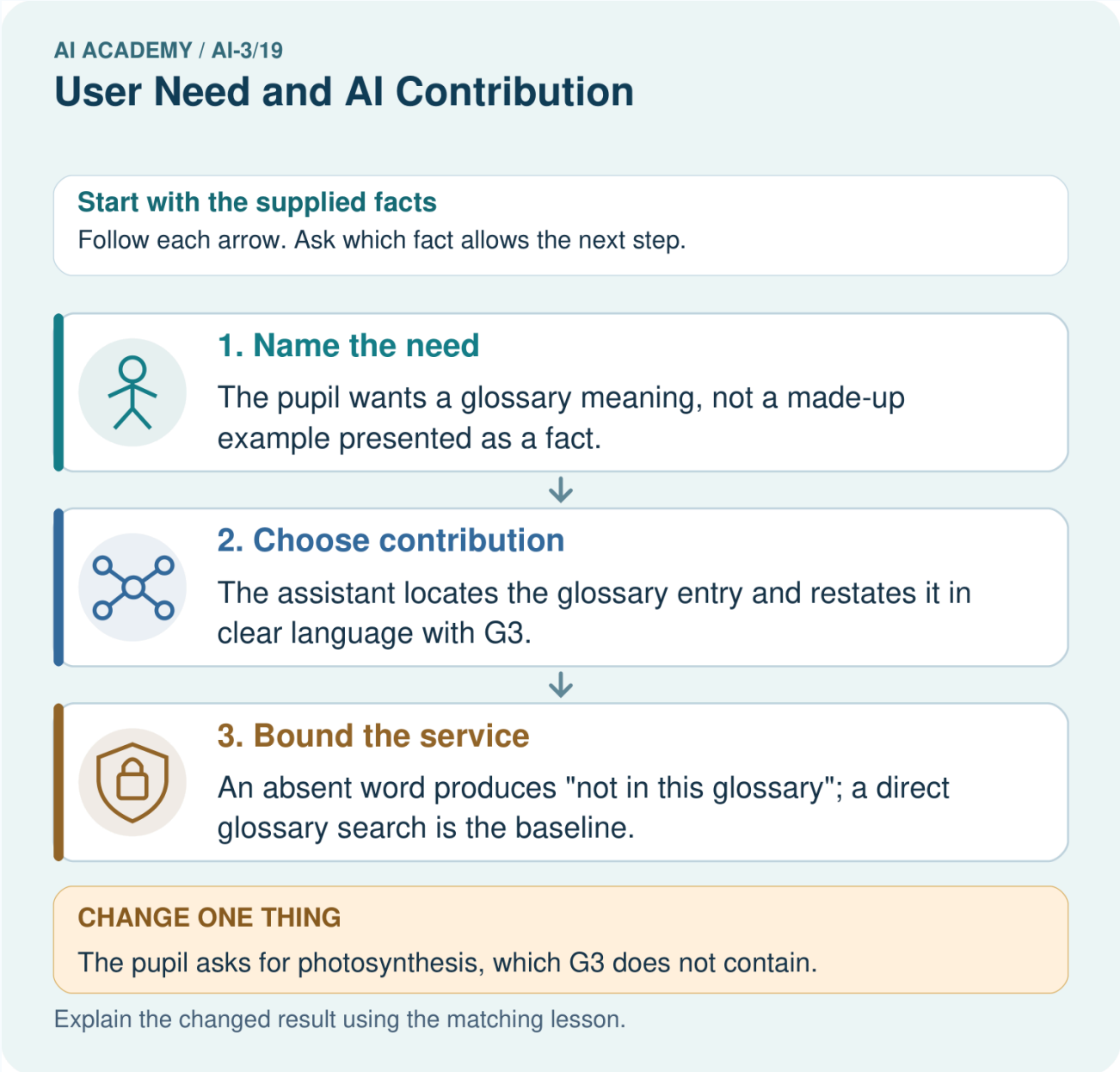
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: User Need and AI Contribution

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a feature mistaken for a benefit

> Evaluate the actual user task rather than the presence of AI.

Repair the friendliness shortcut

Claim to repair: The AI helper sounds friendly, so it must improve learning. Friendly wording is not a learning outcome. First check whether the response preserves the glossary meaning and source. Compare it with direct lookup on the same kind of task, recording errors, time and help. A small classroom comparison can inform a local design choice; it does not establish a general educational benefit.

Keep predicted advantages separate from observed results.

Design a revealing follow-up

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Adding AI automatically makes a product more useful. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

MAIN LESSON · SECTION 8

Practise a bounded corpus plan with less help

> Use room-assistant planning here; reserve the equipment assistant for the first independent attempt.

The plan-versus-performance limit

A complete capstone plan is not **evidence** that the implementation meets it; evaluate after building.

Complete the room-corpus W2

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Capstone **requirement**s and corpus”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Repair an all-future-cases claim

- Use the worked case for Capstone **requirement**s and corpus. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if **need**ed.

- Answer on your own before discussion. If you are unsure, say which step you **need** help with.

Check your reasoning

Explain the glossary baseline

- What non-AI baseline would reveal whether this helper adds value? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Explain the picture

My explanation using a pictured step and the given facts

Keep the independent task separate

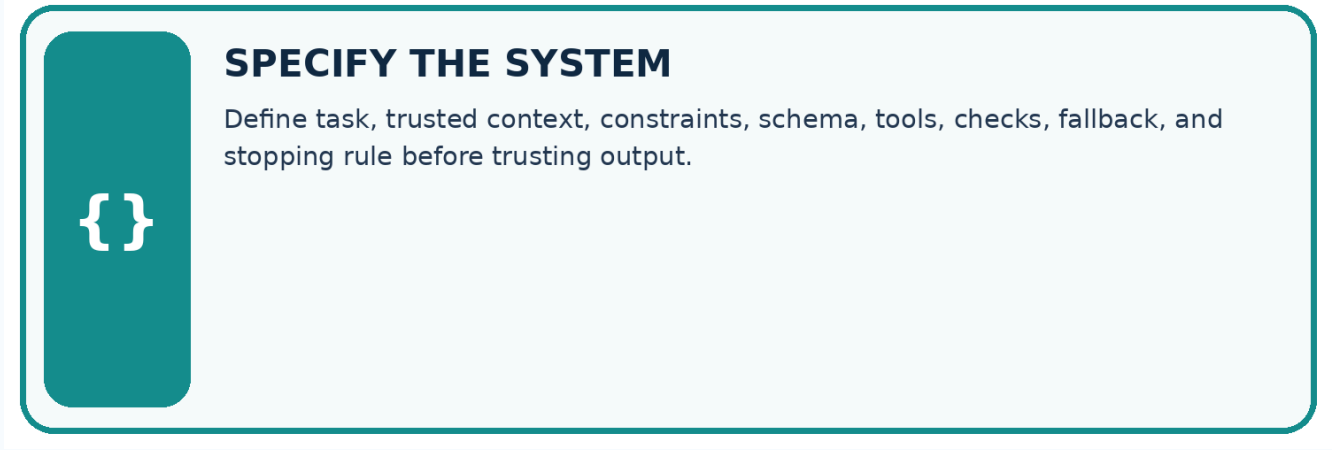
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

MAIN LESSON · SECTION 9

Compare two complete designs for one supplied need

An interview is optional; the fictional learner card makes the exercise self-contained.

All supplied user and design facts

Supplied fictional learner card	Complete task facts
Need statement	I can find a word slowly, but the helper sometimes gives facts outside our approved glossary. I want the approved meaning with its location.
Approved evidence	G3-v1 contains only habitat = the place an organism lives. No photosynthesis entry is supplied.
Design A · Baseline	Direct glossary search displays the entry unchanged, or not in this glossary.
Design B · Proposal	Search G3-v1, restate only its supported meaning, attach source ID; absent entry gives not in this glossary.
Resource constraint	No model/service cost or timing measurements are supplied. Record these as unmeasured rather than inventing a cheaper or faster winner.

Write, sketch, compare and plan a fair check

1. Use the supplied learner card if no interview is available. Write a need statement and sketch A/B flows. For habitat, predict a supported meaning and source; for photosynthesis, predict the absent-entry fallback in both designs. Compare source faithfulness, successful task completion, time, effort, cost and accessibility, marking all unmeasured values unknown.

   Plan a comparison with fresh matched tasks and record prior exposure and support; do not reuse a memorised example as independent evidence. If interviewing a willing classmate, record their actual words separately from this fictional card and avoid personal information. A sketch is not a user trial. Defend whether a model has a necessary role; choosing direct search is acceptable.

   Keep the equipment-assistant assessment withheld.

Keep predictions and observations separate

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____

> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the supplied fictional learner card, or a separately recorded optional classmate interview, to define one **need**. Compare direct lookup with a bounded grounded helper; record source support, fallback and unknown costs or benefits without inventing user-test results.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: scope the fresh equipment assistant

> Choose the evidence boundary and a simpler baseline before features.

Independent need and supplied scope

Student equipment assistant has public inventory I, restricted purchase ledger L and public booking notice B. Define eligible corpus, baseline and a missing-**evidence** test.

Plan the equipment assistant using only the stated public/restricted labels and question. No inventory quantities, booking decisions or live trial results are supplied; this assessment asks for a corpus boundary, baseline and missing-evidence test rather than a fabricated booking answer.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Week 19 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A student **need**s the exact due date from a course notice. Should the design start by generating a plausible date or locating the notice? Explain the **AI contribution**.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Justify exactly which Study Buddy stage uses AI and which stages remain deterministic.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Students cannot find definitions in a 20-page handout. Compare an index with a grounded assistant. Define a success measure and a failure case for both.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project's reason for using AI explicit

> Every proposed feature needs a task and an evaluation plan.

This week's contribution

For the Verified Study Buddy, write the learner's specific task, current difficulty, non-AI baseline, permitted corpus, proposed AI contribution and fallback. Separate deterministic lookup and validation from generated wording. Define what evidence would justify the added feature. Keep features that fail to add value out of the current scope; a smaller reliable tool can better serve the need.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Retain the checkpoint and human choice

THIS WEEK'S CHECKPOINT Justify exactly which Study Buddy stage uses AI and which stages remain deterministic.

SYSTEM MAP Learner **need** → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

Write or draw your reasoning and evidence.

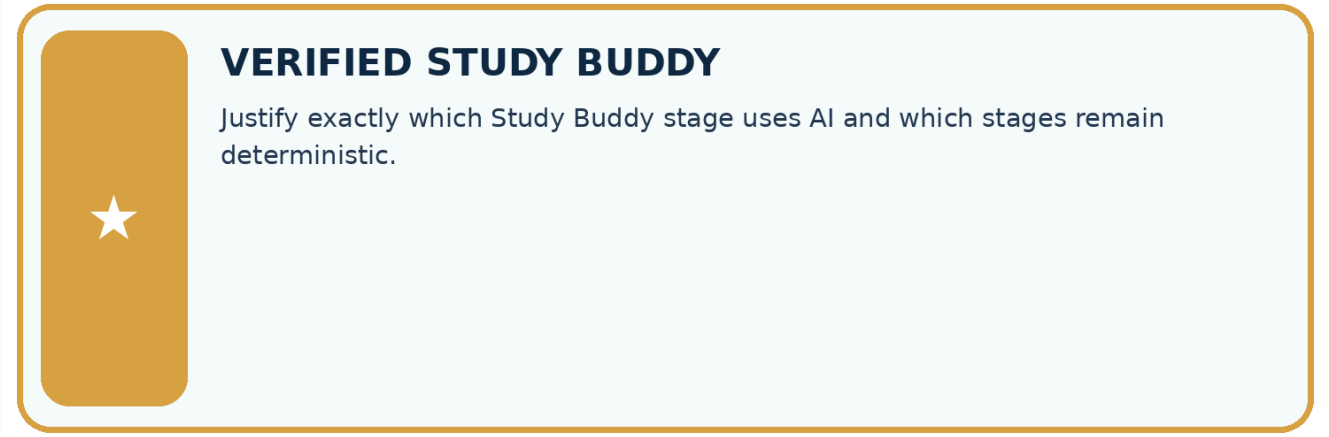
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse fluent **output** with verified truth.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

Finish: name the need, baseline and evidence of value

> A plan must later be tested against the actual user task.

Explain without notes

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of capstone **requirement**s and corpus

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
user	The person who interacts with or relies on a system.	
need	A problem or desired outcome for the intended user.	
AI contribution	The specific part of a task an AI system is intended to help perform.	
simpler alternative	A less complex approach that may meet the same user need with fewer resources or risks.	

Terms for the connected explanation

Word	Meaning
Requirement	A declared user-visible condition the system must satisfy
Corpus manifest	An inventory of source IDs, versions and access roles

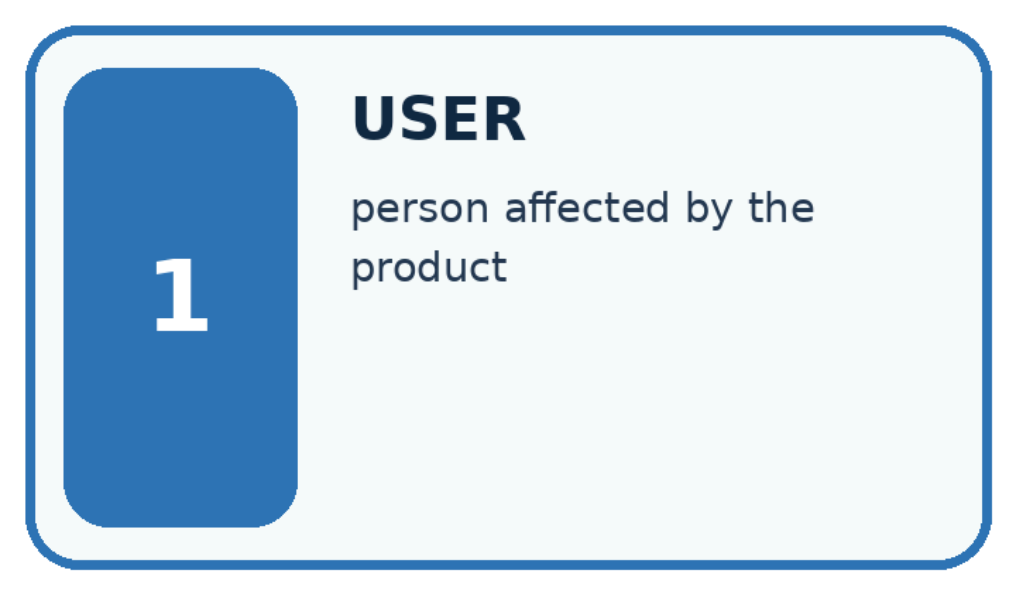
Word	Meaning
Baseline procedure	A credible reference process on the same task

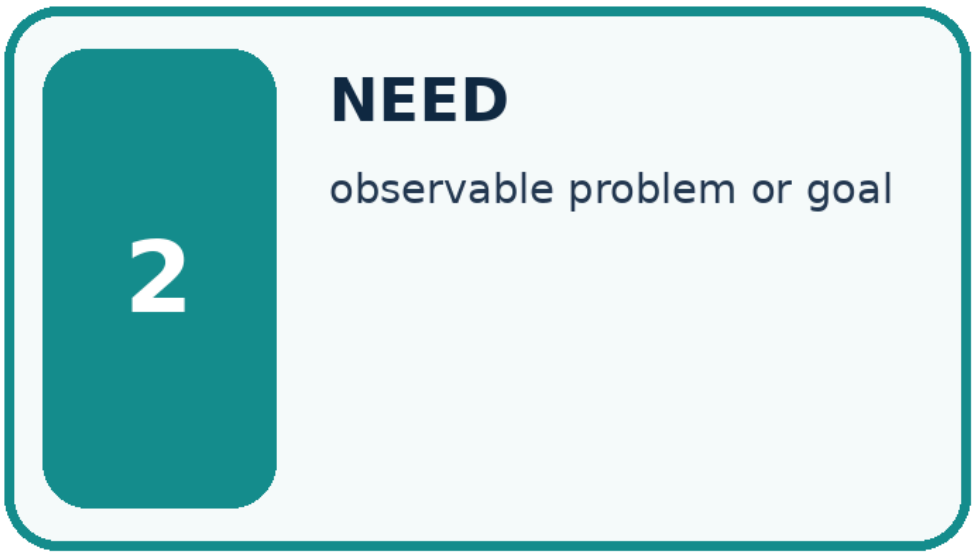
▼ Earlier wording, alternatives and picture transcripts

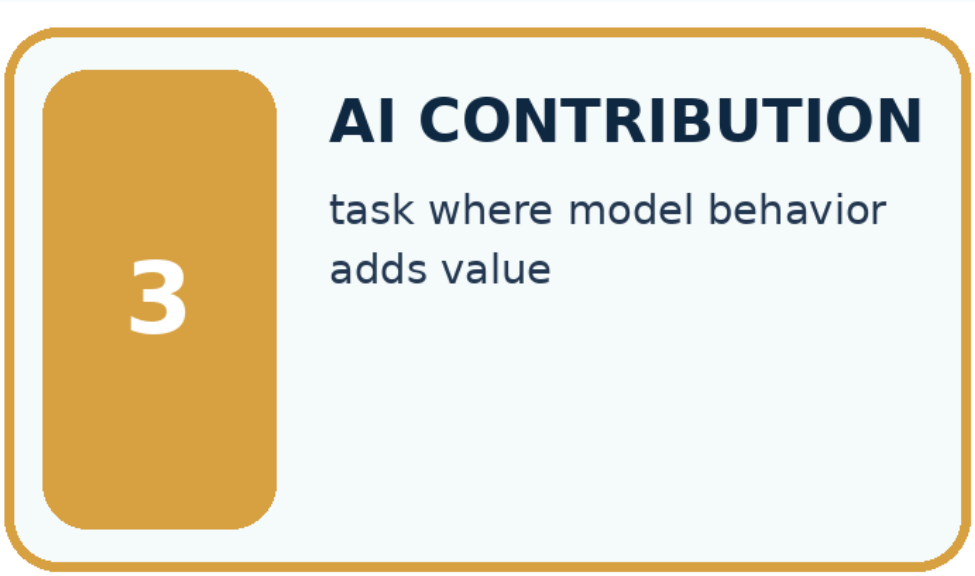
VOCABULARY LAB

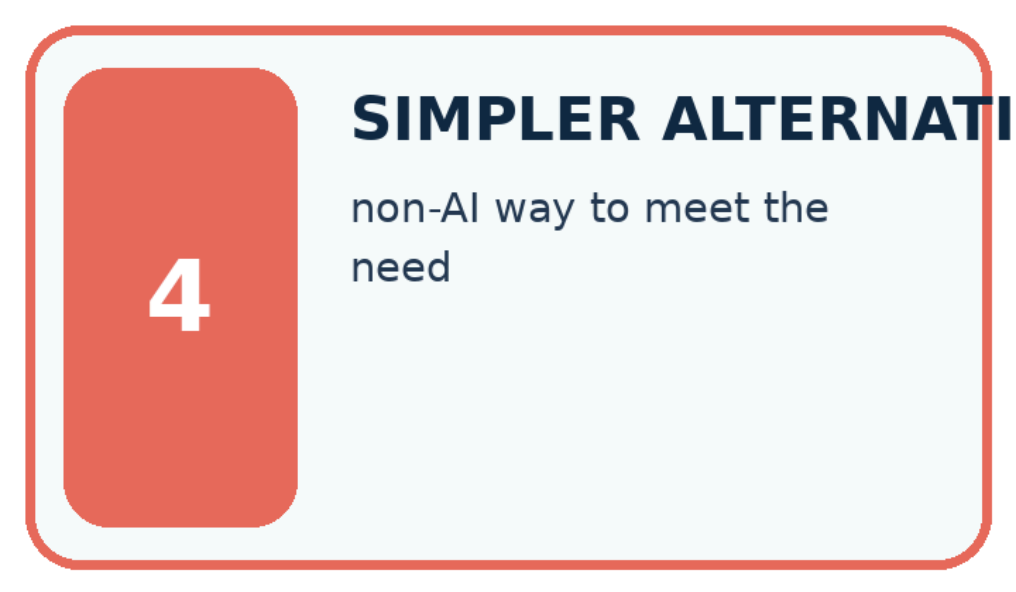
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
NEED
observable problem or goal


3
AI CONTRIBUTION
task where model behavior adds value



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

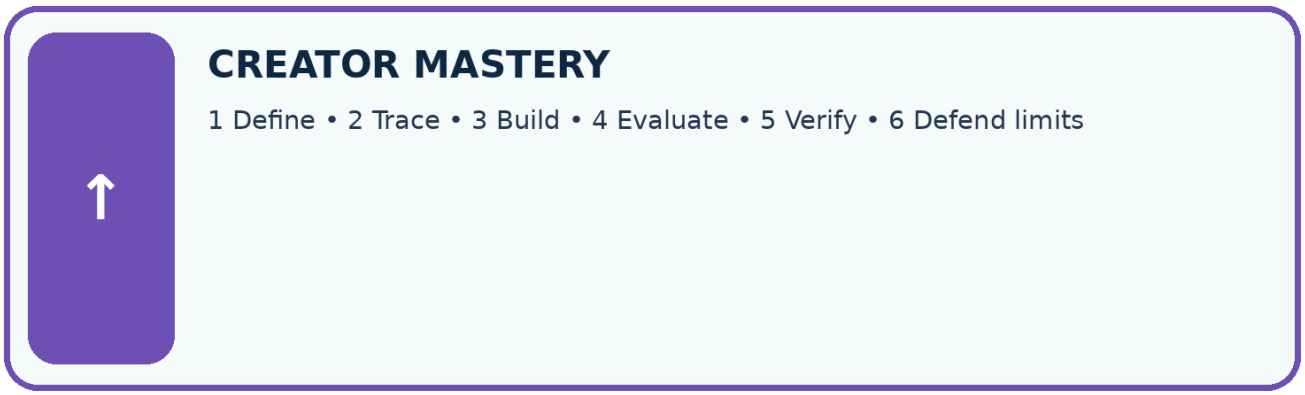
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Scope the equipment assistant independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation **need**ing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

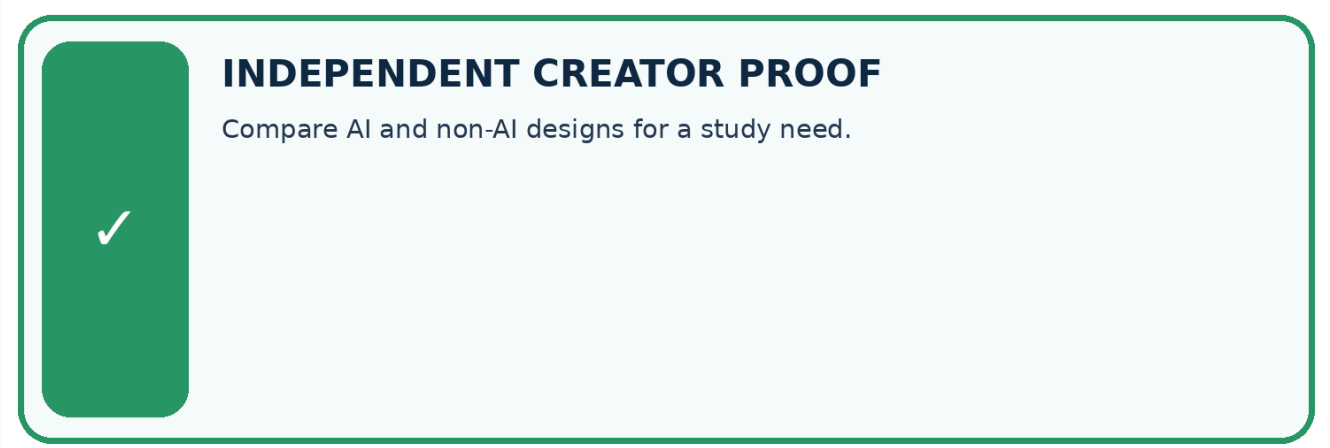
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Scope the equipment assistant independently is the selected independent case.



EXPLANATION CHECK Require eligible evidence, a non-AI baseline, missing-evidence behavior and no invented performance result.

Write or draw your reasoning and evidence.

▼ **Extra practice: Choose a narrow, testable contribution**

REFERENCE · SECTION 13

Extra practice: Choose a narrow, testable contribution

> Try the question before opening its worked explanation.

Read the supplied case and question

Learners struggle to locate definitions in an approved chapter; a proposed chatbot answers everything.

Narrow the **need** and **AI contribution**.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Help locate and explain chapter definitions with references and unsupported-question handling. Compare a simple index or search first.

Evidence or stage	Value or result
Need	Find chapter definitions
Simple baseline	Index or search
AI help	Explain with chapter references
Limit	Handle unsupported questions

▼ Earlier wording, alternatives and picture transcripts

Week 19 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Choose a narrow, testable contribution

Learners struggle to locate definitions in an approved chapter; a proposed chatbot answers everything.

Narrow the **need** and **AI contribution**.

Help locate and explain chapter definitions with references and unsupported-question handling. Compare a simple index or search first.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Explain the mechanism and terminology of capstone requirements and corpus	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Define a useful reading helper before choosing its AI features.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- Define a useful reading helper before choosing its AI features.
- Which labelled step matches this detail? An absent word produces “not in this glossary”; a direct glossary search is the baseline.
- 1. Bound the service
- 2. Name the **need**
- 3. Choose contribution
- Hint if **need**ed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.
- After your attempt, compare with the supported explanation: Bound the service An absent word produces “not in this glossary”; a direct glossary search is the baseline. Read the picture in order.

  An absent word produces “not in this glossary”; a direct glossary search is the baseline. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check the first explanation

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: The pupil asks for photosynthesis, which G3 does not contain. Which conclusion is supported by the supplied facts?
- 1. The assistant reports that the definition is unavailable in its approved glossary.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if **need**ed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.
- After your attempt, compare with the supported explanation: The assistant reports that the definition is unavailable in its approved glossary. A helpful interface cannot expand its authorised **evidence** by guessing.

My selected conclusion, deciding fact and explanation

▼ Check the changed case

Repair a missing building block

Repair and save your work

Review Retrieval Augmented Generation (ai-3/18). Goal: Explain the **mechanism** and terminology of building a grounded answer. Short practice: Which part of the source supports a two-minute melting claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Verify Numbers and Dates (ai-3/16). Goal: Verify a numeric claim using its source, denominator and units. Short practice: Which part can be corrected immediately from existing **evidence**, and which **need**s follow-up?. Return to this week and explain what you changed.

Review Multimodal **Evidence** (ai-3/13). Goal: Keep visible **evidence**, hidden contents and caption claims separate. Short practice: Why were both “exactly three” and “exactly five” unsupported before opening the box?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

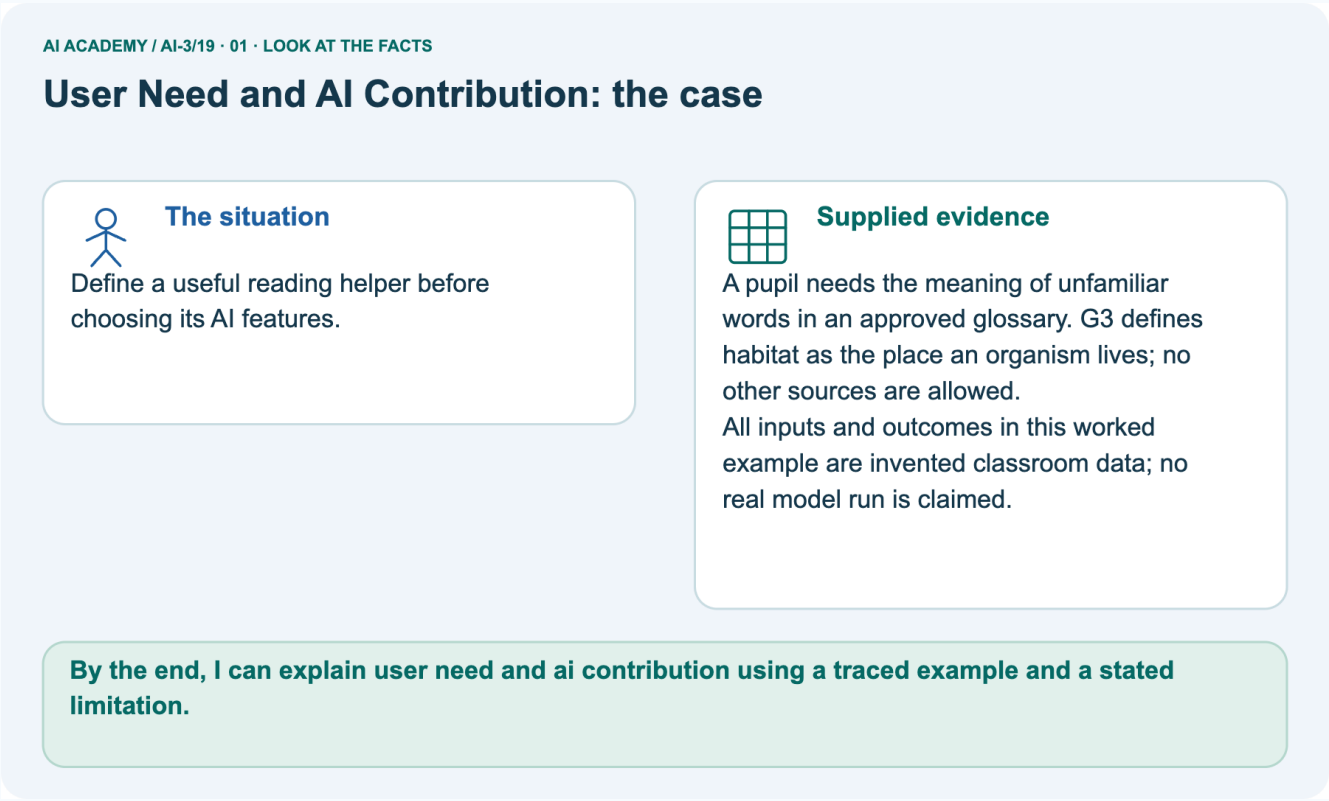
Keep your completed work

▼ Reference: the six-picture journey

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. User Need and AI Contribution: the case



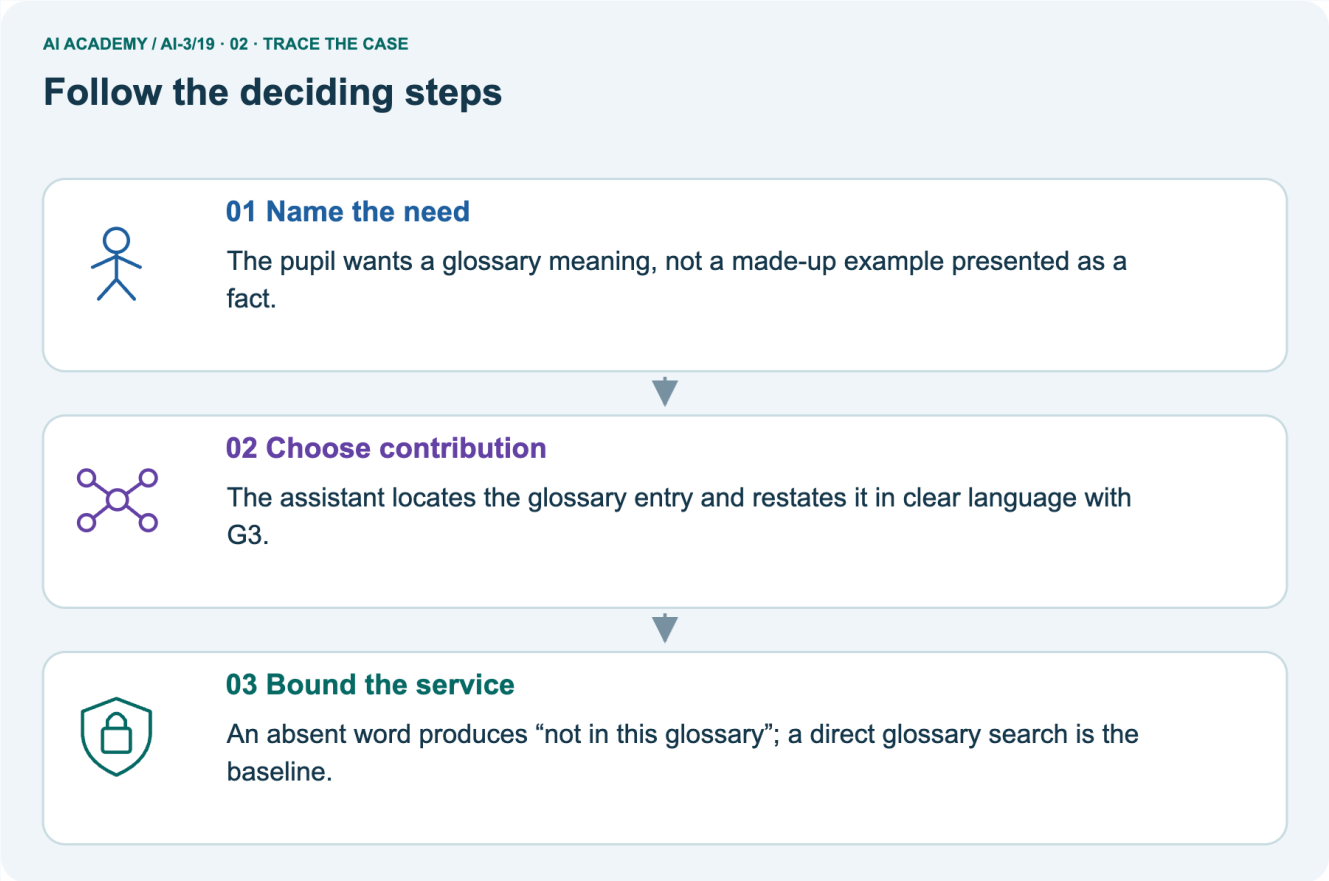
Follow the labelled picture with the matching explanation. Open the vector diagram

Define a useful reading helper before choosing its AI features. Supplied facts: A pupil **need**s the meaning of unfamiliar words in an approved glossary.

G3 defines habitat as the place an organism lives; no other sources are allowed. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

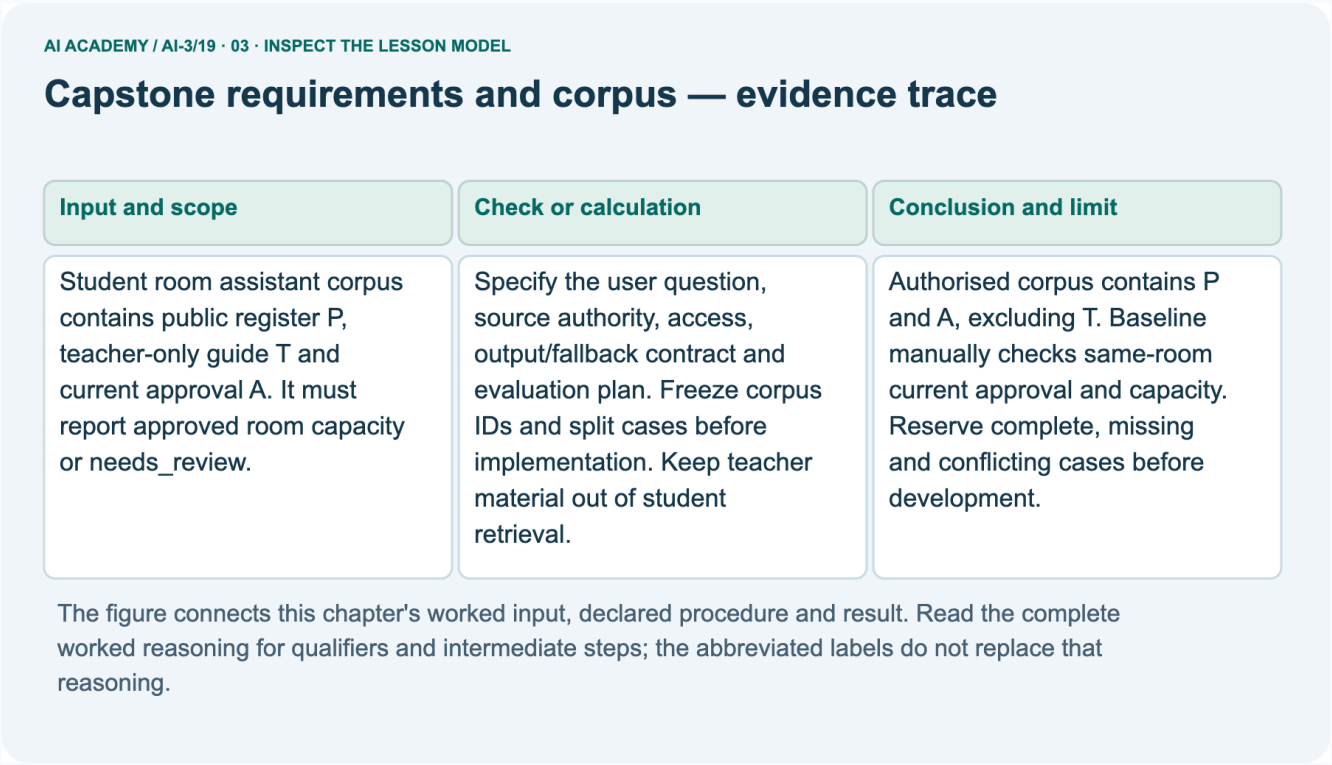


Follow the labelled picture with the matching explanation. Open the vector diagram

Name the **need**: The pupil wants a glossary meaning, not a made-up example presented as a fact. → Choose contribution: The assistant locates the glossary entry and restates it in clear language with G3. → Bound the service: An absent word produces “not in this glossary”; a direct glossary search is the baseline.

Point to each numbered stage and explain which recorded **input** it uses.

03. Capstone requirements and corpus — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone **requirement**s and corpus — **evidence** trace. Student room assistant corpus contains public register P, teacher-only guide T and current approval A.

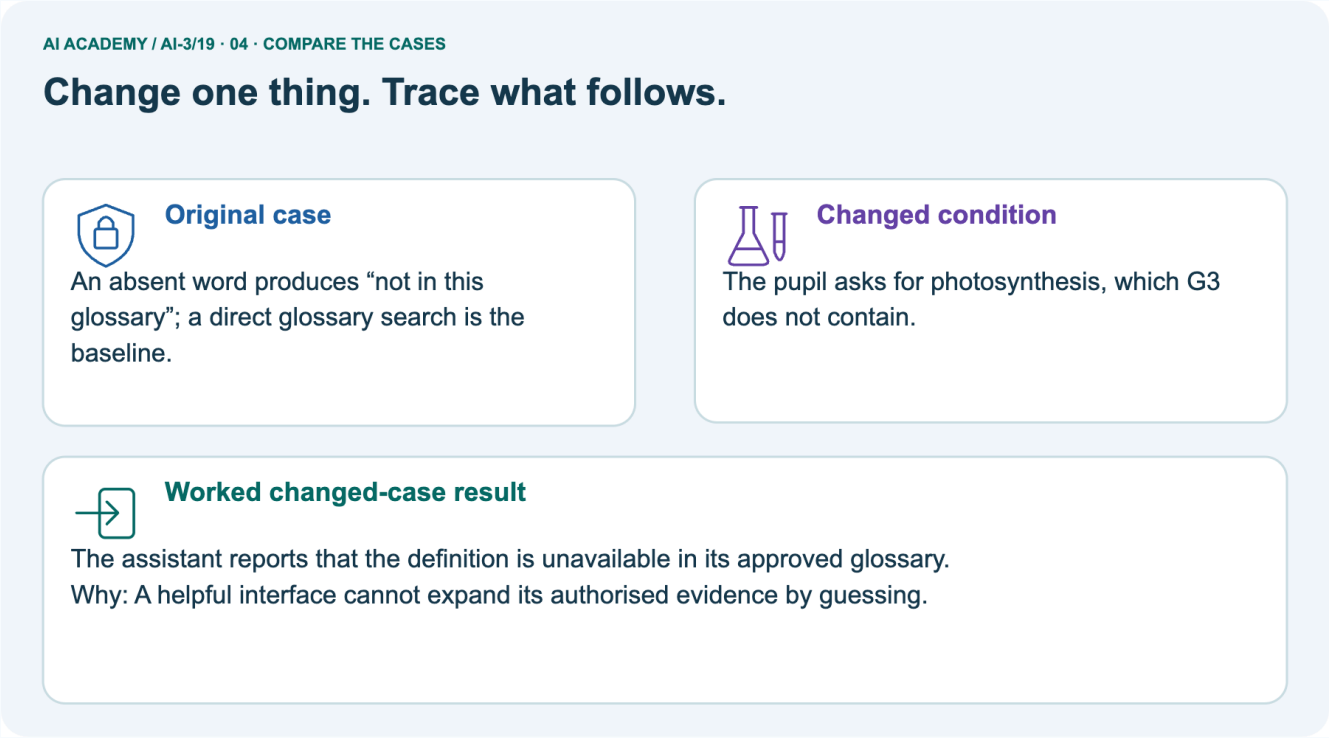
It must report approved room capacity or **need**s_review. | Specify the **user** question, source authority, access, **output**/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation.

Keep teacher material out of student retrieval. | Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity.

Reserve complete, missing and conflicting cases before development.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



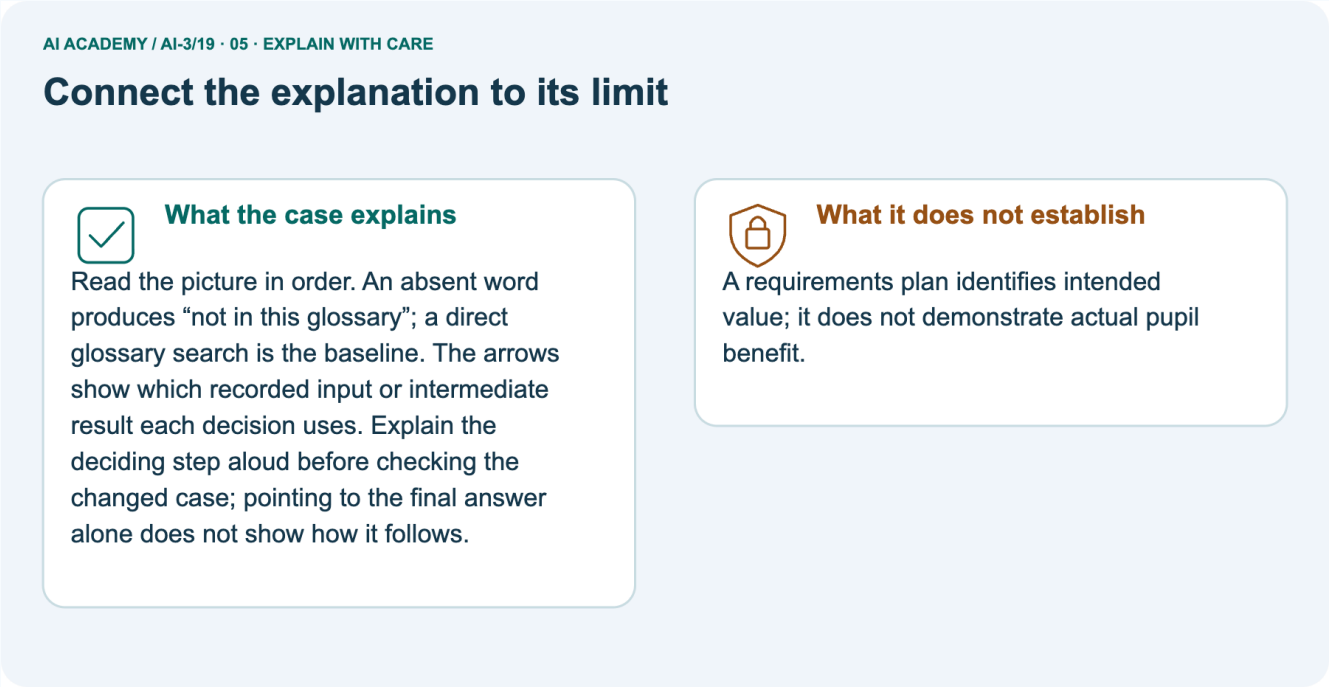
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: An absent word produces “not in this glossary”; a direct glossary search is the baseline. Changed condition: The pupil asks for photosynthesis, which G3 does not contain.

Expected result: The assistant reports that the definition is unavailable in its approved glossary. Reason: A helpful interface cannot expand its authorised **evidence** by guessing.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

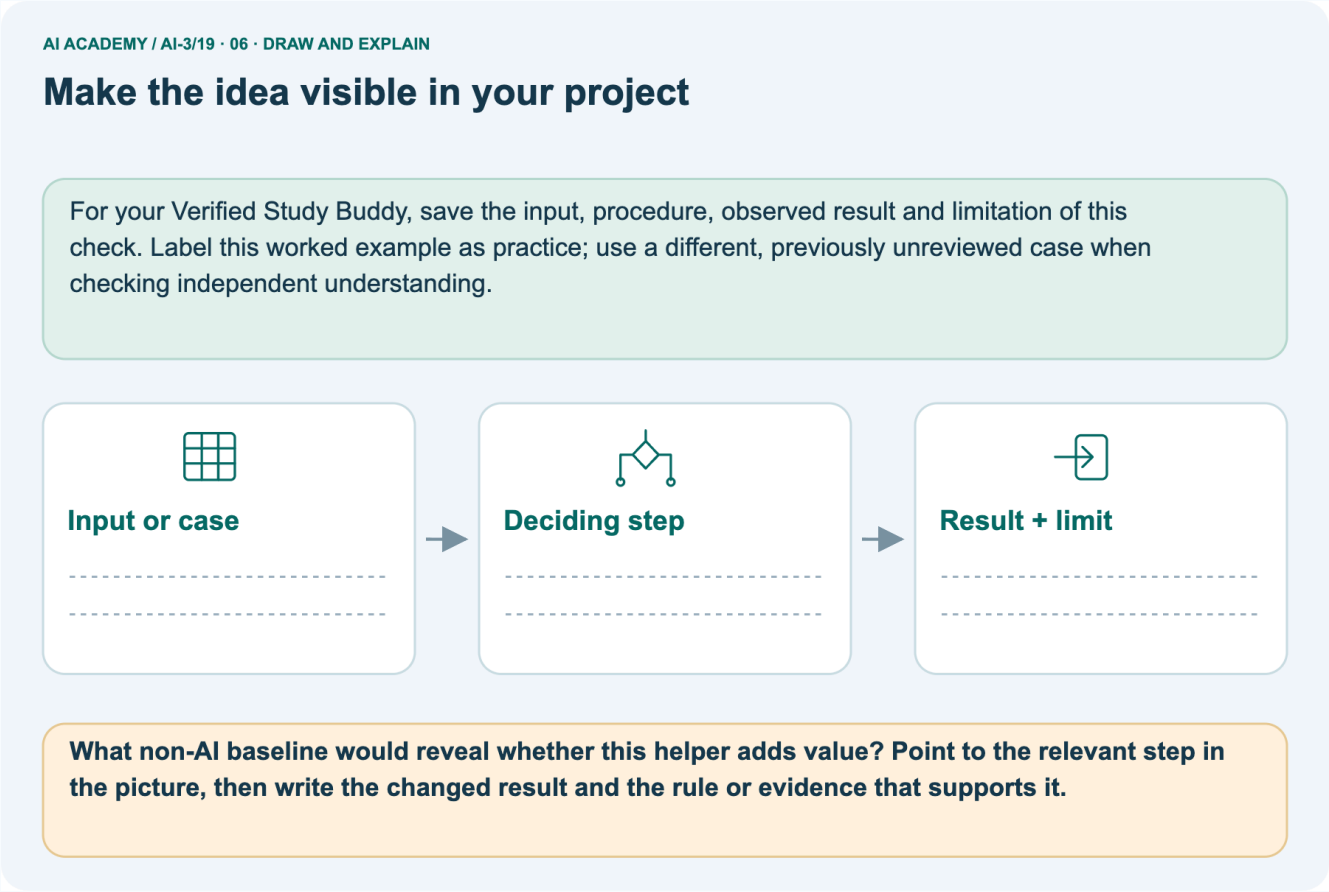
Read the picture in order. An absent word produces “not in this glossary”; a direct glossary search is the baseline. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A **requirement**s plan identifies intended value; it does not demonstrate actual pupil benefit.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What non-AI baseline would reveal whether this helper adds value? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. User Need and AI Contribution: the case

02. Follow the deciding steps

03. Capstone requirements and corpus — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.